

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	AI based crop recommendation
Maximum Marks	5 Marks

Functional Features Overview:

This document describes the functional features implemented in the OptiCrop – AI-Based Crop Recommendation System. Each feature contributes to recommending the most suitable crop based on soil nutrients (N, P, K) and environmental conditions such as temperature, humidity, pH, and rainfall. The system provides a simple and interactive web interface for users to input data and view predictions.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Input Form	Allows users to enter soil and climate parameters for prediction.	Frontend	Done	
2	Data Preprocessing	Cleans and prepares dataset for training and prediction.	Backend	Done	
3	Machine Learning Model	Predicts the most suitable crop using trained model.	ML Model	Done	
4	Crop Prediction Result	Displays recommended crop to the user.	Backend	Done	
5	Responsive Web Interface	User-friendly interface for easy interaction.	Frontend	Done	
6	Input Validation	Ensures correct numeric values are entered.	Backend	Done	
7	Model Integration With Flask	Connects ML model with web application.	Flask Backend	Done	
8	GitHub Repository Management	Stores and maintains project source code.	Version Control	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Crop Input Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Input Validation, Prediction Logic
3	Machine Learning	1	Crop Recommendation Model
4	API / Integration	1	Flask + ML Model Integration
5	Data Storage / Assets	1	Dataset (CSV), Trained Model (.pkl)